



# 4

## Exploring Magnets

### NCF features

- Class Discussions
- Higher Order Thinking
- Activity-based Learning
- Problem-solving
- Inquiry-based Learning
- Collaboration
- Digital Literacy
- Experiential Learning

### WHAT IS A MAGNET?

#### THINK AND DISCUSS! Reflective thinking

Have you ever played with magnets? Recall the shapes of the magnets you have seen. Which objects did the magnets attract? Which objects did they not attract?

More than 2500 years ago, the Greeks discovered a certain type of black stone that had the power of attracting pieces of iron. The story goes that Magnes, a shepherd, had a stick with a small iron piece attached at one end. One day his stick got stuck to a black rock. It was discovered that the rock had the property of attracting iron. The Greeks named the rock magnetite.

It was also discovered that if a piece of that stone was suspended freely, it always came to rest pointing in the north-south direction. This property of the stone was found to be very useful in determining directions: It was, therefore, named lodestone, which means 'leading stone'. The stones were later given the name magnets.

An object that has the property of attracting iron and a few other materials is called a magnet. The property itself is called magnetism. The force that the magnet exerts is called magnetic force.

If you have played with a magnet, you would have observed that it can attract an iron nail from a distance. Thus, magnetic force acts from a distance even when there is no contact (Fig. 4.1).



Fig. 4.1 Magnetic force acts from a distance.

### Types of magnets

Lodestone is a naturally occurring magnet. The magnets present in pencil boxes, magnetic stickers and in toys are artificial magnets. They are made from iron and some other materials.

#### IN THIS CHAPTER

WHAT IS A MAGNET? ♦ POLES OF A MAGNET ♦ FINDING DIRECTIONS WITH A MAGNET  
♦ THE EARTH'S MAGNETISM ♦ MAKING MAGNETS ♦ INFLUENCE OF A MAGNET ♦ USES OF MAGNETS

Artificial magnets can be of various shapes (Fig. 4.2). A school laboratory usually has **bar magnets** and **horseshoe magnets**.

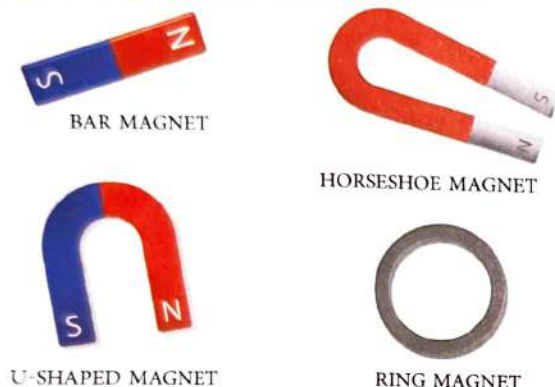


Fig. 4.2 Some common types of magnets

## MAGNETIC AND NON-MAGNETIC MATERIALS

Let us do Activity 1 for some objects.

Materials that are attracted by a magnet are called **magnetic materials**. Examples of magnetic materials are **iron, nickel and cobalt**.

Mixtures of these with other metals and non-metals are also attracted by a magnet. Materials which are not attracted by a magnet are called **non-magnetic materials**. Examples are **gold, silver, copper, aluminium, wood, plastic and rubber**.

## POLES OF A MAGNET

Let us now see which regions in a magnet have stronger magnetism.



### ACTIVITY 2 (Experimental investigation)

1. Spread some iron filings on a sheet of paper.
2. Roll a bar magnet in the filings and then lift it up. What do you observe?
3. Repeat with a horseshoe magnet. What do you observe?

Do the iron filings stick to the magnet uniformly along its length? Or do they stick to certain portions of the magnet?



### ACTIVITY 1 (Experimental investigation)

Collect a few objects such as coins, metal spoons, wood, rubber, plastic, nails and pins.

Predict whether each object will be attracted by a magnet or not. Now, bring a magnet close to each object. Observe which objects are attracted by the magnet.

Was your prediction correct for each object? Fill in Table 4.1.

Table 4.1 Magnetic and non-magnetic materials

|    | OBJECT | MADE OF | WHETHER ATTRACTED BY MAGNET |             |
|----|--------|---------|-----------------------------|-------------|
|    |        |         | MY PREDICTION               | OBSERVATION |
| 1. |        |         |                             |             |
| 2. |        |         |                             |             |
| 3. |        |         |                             |             |
| 4. |        |         |                             |             |

You will find that things made of iron and steel (steel is made by mixing iron with other metals and non-metals) are attracted by the magnet. Things made of other materials such as wood, plastic, brass, copper, gold and silver are not attracted by the magnet.

You will find that most of the iron filings stick to the magnets at the ends. There are fewer iron filings in between and almost none at the centre (Fig. 4.3). Thus, in a bar or a horseshoe magnet, the regions of strongest magnetism are near the ends.



Fig. 4.3 Poles of magnet

The ends of a magnet, where the magnetic force is the strongest, are known as the **poles** of the magnet.



### THINK AND DISCUSS!

Critical thinking



Fig. 4.4

You have seen that a bar magnet and a horseshoe magnet have poles at the end.

**H.W** What about a ring magnet? Where do you think its poles could be?

## FINDING DIRECTIONS WITH A MAGNET

Let us now see what happens when a magnet is suspended freely.



### ACTIVITY 3 (Experimental investigation)

1. Place the magnet in a stirrup made of paper and suspend the stirrup using a thread (Fig. 4.5).
2. Make sure that there is no magnet or magnetic material placed nearby.

3. Allow the magnet to come to rest, and note the direction in which it comes to rest.
4. Now rotate the magnet by giving one of its ends a gentle push and wait for it to come to rest again.
5. What do you observe?
6. Now repeat the activity using a small iron bar in place of a magnet. What do you observe?

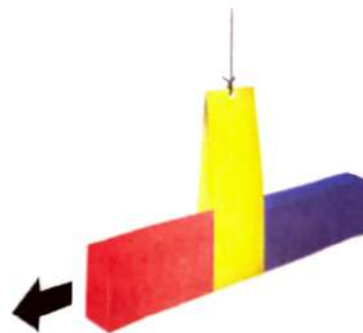


Fig. 4.5 A freely suspended magnet

You will observe that the freely suspended magnet always comes to rest in the north-south direction, with one end always pointing towards the geographic north of the earth. This is called the direction-finding property of a magnet. You will also observe that an iron bar comes to rest in any direction.

The end of a magnet that points towards the geographic north of the earth is called the north-seeking pole or the north pole of the magnet.

The end of a magnet that points towards the geographic south of the earth is called the south-seeking pole or the south pole of the magnet.

## The magnetic compass

A magnetic compass uses the direction-finding property of a magnet to find direction. It is especially useful for sailors. It consists

a small circular box of non-magnetic material with a magnetized needle pivoted at a point in the centre of the box. The needle is free to rotate about that point. Its north pole is usually marked red. The needle always comes to rest in the north-south direction. After the needle has come to rest, the compass is gently rotated until the north and south marked on the dial are aligned with the needle (Fig. 4.6). The markings on the dial now indicate the directions.



Fig. 4.6 A magnetic compass



#### ACTIVITY 4 (Model making)

##### To make a compass

1. Magnetize an iron sewing needle with the help of a bar magnet, as explained in Activity 7.
2. Pass the magnetized needle horizontally through a cork (Fig. 4.7).
3. Float the cork in a bowl of water such that the needle always remains above the water level.



Fig. 4.7 A floating magnetic compass

Your compass is ready. Test your compass by gently rotating the cork a few times. Does the needle always come to rest in the north-south direction?

## Can magnetic poles be isolated?



### THINK AND DISCUSS! Reflective thinking

Every magnet has two poles—a north pole and a south pole. What do you think will happen if you cut the magnet into two? Do you think that the north pole will get separated from the south pole? Or do you think each piece will be a complete magnet with its own north and south pole?

It is found that if we cut a magnet into two, we end up with two separate magnets, each with its own north and south poles. If we further cut these two magnets, we get four separate magnets, each with its own north and south poles. We can continue this process and get smaller and smaller magnets, each with a north and a south pole (Fig. 4.8).

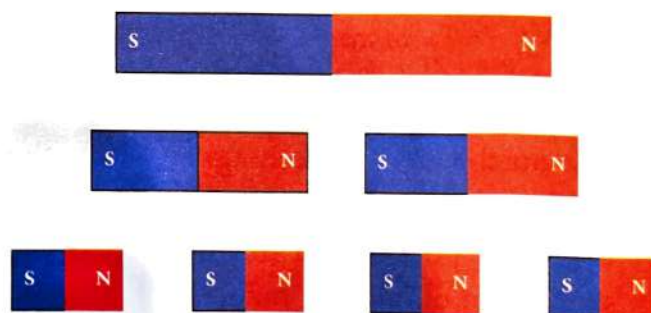


Fig. 4.8 Cutting a magnet

We, therefore, conclude that:

- ❖ poles always exist in pairs, and
- ❖ we can never have an isolated north pole or south pole.

## Attraction and repulsion in a magnet



#### ACTIVITY 5 (Experimental investigation)

1. Suspend a bar magnet freely.
2. Slowly bring the north pole of another bar magnet towards the north pole of the suspended magnet.

3. What do you observe? Do the two poles attract or repel each other?
  4. Now bring the north pole of the second magnet towards the south pole of the suspended magnet.
  5. What do you observe? Do the two poles attract or repel each other?
- Repeat the experiment with the south pole of the second magnet.

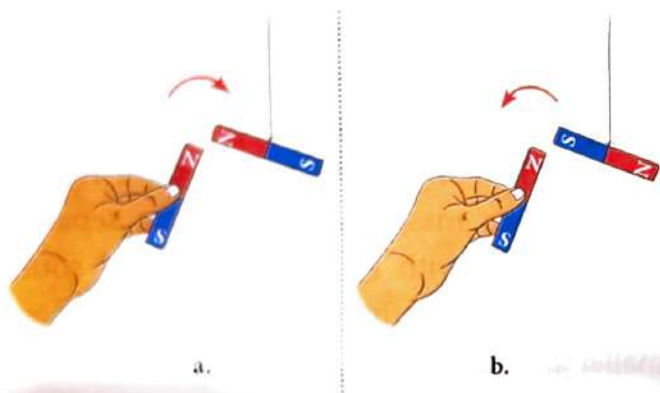


Fig. 4.9 Repulsion and attraction in magnets

You will notice that:

- ❖ The north pole of one magnet **repels** the north pole of another magnet. Similarly, the south pole of one magnet **repels** the south pole of another magnet.
- ❖ The north pole of one magnet **attracts** the south pole of another magnet.

In other words, **unlike poles attract and like poles repel each other.**

- ❖ Repeat Activity 5 by using an iron bar in place of one of the magnets. What do you observe? You will observe that both ends of the iron bar are attracted by both the north and south poles of the magnet.

### Test for magnetism

To test if a metal bar is a magnet or not, do Activity 6.



### ACTIVITY 6 (Experimental investigation)

1. Bring one pole of a magnet close to the ends of a metal bar, one end at a time.
2. If one end is attracted and the other repelled, the metal bar is a magnet. If both ends are attracted, the metal bar is not a magnet.

Thus, **repulsion and not attraction, is a sure test for magnetism.**



### THINK AND DISCUSS!

Logical thinking



- A. The north pole of a bar magnet is slowly brought close to the north pole of a compass needle at rest.
- B. The south pole of a bar magnet is slowly brought close to the north pole of a compass needle at rest.

Fig. 4.10

In which case, A or B, will the compass needle be deflected (no longer point in the north-south direction) as the bar magnet comes near it? Mark the direction of deflection of the north pole of the compass needle with an arrow.

## THE EARTH'S MAGNETISM

Why does a freely suspended magnet always come to rest in the north-south direction? This is because the earth itself behaves like a huge bar magnet with its magnetic poles near the geographical North and South Poles.

The south pole of this bar magnet is near the earth's geographical North Pole, and the north pole of the bar magnet is near the earth's geographical South Pole.

In a freely suspended magnet, therefore, the north pole points towards the geographical North Pole since it is attracted by the earth's magnetic South Pole.

Similarly, the south pole of the suspended magnet is attracted by the earth's magnetic North Pole and, therefore, points towards the geographical South Pole (Fig. 4.11).

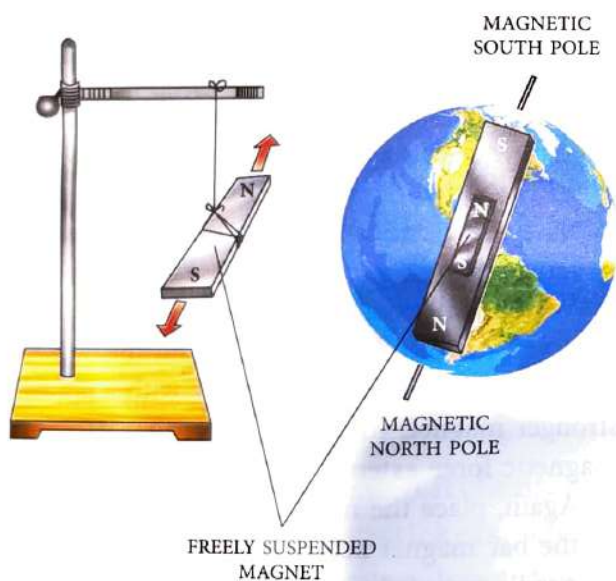


Fig. 4.11 The earth behaves like a huge bar magnet.

### It's a Fact!



The earth's magnetic poles attract some special particles emitted by the sun. When these particles strike molecules in air, coloured light is given out. This display of coloured light is called an **aurora**.



## Oral Questions for Class Discussion

1. Do all parts of a magnet have equal magnetic strength? Justify your answer with reference to a bar magnet and a horseshoe magnet.
2. A freely suspended magnet comes to rest in the north-south direction. Why is that so?
3. If you break a magnet into 10 pieces, how many north and how many south poles would there be in all?
4. a. One end of an iron bar is attracted and the other end is repelled by the north pole of a bar magnet. Can you definitely say that the iron bar is a magnet?  
b. If both ends are attracted by the north pole, what would be your conclusion?

## MAKING MAGNETS

Magnetic materials such as a piece of iron can be magnetized by a magnet in the following way.



### ACTIVITY 7 (Model making)

1. Place an iron needle or a bar on a table.
2. Hold it down firmly and stroke it in one direction only with one pole of a bar magnet (Fig. 4.12). After you reach the other end, lift the magnet high and bring it back to the first end.
3. Repeat 30–40 times.
4. Now test the needle to see if it has become a magnet. Suspend it freely to see which end has become the north pole.

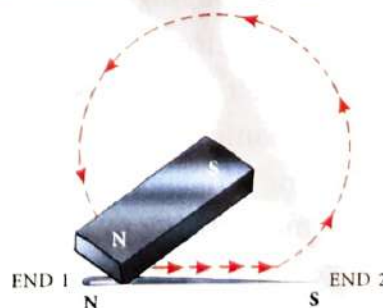


Fig. 4.12

You will find that if you stroke with the north pole of the magnet, the end of the iron bar from which the stroking is started (end 1) becomes the north pole. The other end (end 2) becomes the south pole. If you stroke with the south pole, the poles in the iron needle will be reversed. This is known as the **single touch method** of making magnets.

**Using electric current:** The above method of magnetization is, however, no longer used. These days all magnets are made with the help of electric current. You will read about this in higher classes.

### Care and storage of magnets

- ❖ Magnets tend to become weaker after some time if their poles are left free. This is called **self-demagnetization**. To avoid this, keep bar magnets in pairs separated by a piece of wood, with unlike poles alongside each other. Place a pair of iron pieces, called **keepers**, across both ends. A horseshoe magnet needs only one keeper across its poles.

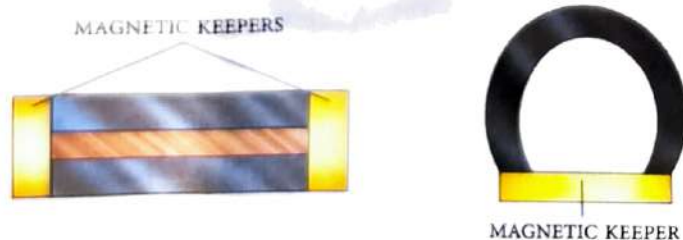


Fig. 4.13 Storing a magnet

- ❖ Take care not to drop a magnet—a magnet loses its magnetism if it is dropped from a height or hammered.
- ❖ Avoid heating a magnet as this also results in loss of magnetism.
- ❖ Do not keep powerful magnets near TVs, computers, mobile phones, and remote controls.

## INFLUENCE OF A MAGNET

We have seen that magnetic force acts from a distance. How far does the influence of the magnetic force extend? Can the magnetic force act through non-magnetic materials?



### ACTIVITY 8 (Experimental investigation)

1. Place a compass needle on a table.
2. Place the north or south pole of a bar magnet near the needle so that the needle gets deflected.
3. Slowly move the magnet away from the compass in any direction. Does the deflection of the needle reduce as the magnet moves further and further away from it?
4. Note the distance at which the deflection cannot be observed at all.

We conclude that the magnetic force is observed in a limited region around the magnet. If you repeat the activity with a stronger magnet, you will observe that its magnetic force extends to a greater distance.

5. Again, place the north or south pole of the bar magnet near the needle so that the needle gets deflected.
6. Now, place a sheet of plastic between the magnet and the compass. Observe if there is any change in the deflection of the needle.
7. Repeat with sheets of other non-magnetic materials such as wood, cardboard and glass.



Fig. 4.14 Magnetism can act through plastic.

You will observe that there is no noticeable change in the deflection with any of these materials. We conclude that the magnetism can act through non-magnetic materials.

Activity 8 shows that:

- ❖ There is a region all around a magnet where its magnetic force can be felt.
- ❖ The stronger the magnet, the greater is this region.
- ❖ The magnetic force reduces with increase in distance.
- ❖ Presence of a non-magnetic material does not affect the magnetic force.

### **THINK AND DISCUSS!** Reflective thinking

Do you have a magnet-type influence around you? Is it possible for you to kick a football from a distance?

## USES OF MAGNETS

Magnets find a large number of applications. Some of these are as follows:

- ❖ In a compass for finding directions.
- ❖ In refrigerator door stickers, closing mechanism of refrigerator doors, bulletin boards and toys.
- ❖ In electric motors used in fans, washing machines, mixer-grinders etc.
- ❖ In bicycle and automobile dynamos for making electricity.
- ❖ In speakers, microphones, picture tubes of televisions and computer monitors.
- ❖ In computer hard disks to store information.



a. Hard disk



b. Credit card

Fig. 4.15 Some uses of magnets

- ❖ In ATM cards and credit cards where information is stored in a magnetic strip.
- ❖ For picking up iron pieces from waste.

## FUN WITH MAGNETS

Here are some fun activities to try out with magnets.

### **ACTIVITY 9** (Art integration)

#### Dancing snake

1. Tie a steel paper clip at the end of a string, about 30 cm long (Fig. 4.16a).
2. Cut out two similar heads of a cobra snake from paper. Glue the paper clip in the centre of one of them (Fig. 4.16b). Paste the other head on top. Now the paper clip cannot be seen.
3. Secure the other end of the string to a table using tape, as shown in Figure 4.16c.
4. Cover a strong bar magnet with paper to make it look like a magician's wand.
5. When you make your magnetic wand hover over the cobra, it lifts up its head. Make it dance by moving the wand (Fig. 4.16c).

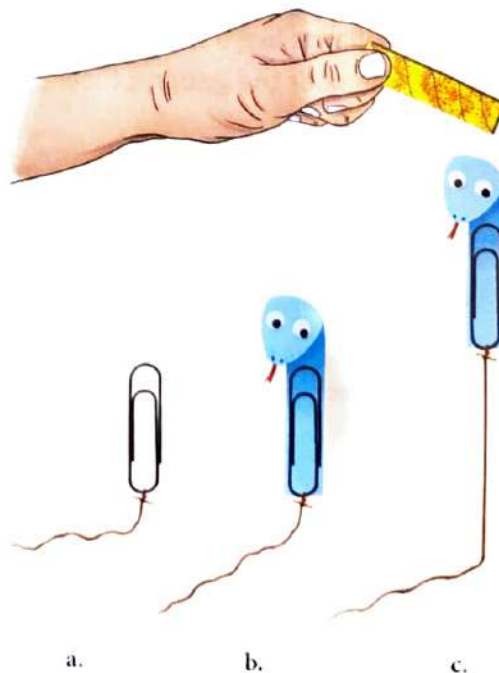


Fig. 4.16



## ACTIVITY 10 (Art integration)

### Painting with magnets

1. Put a few drops of paint of different colours on a paper plate. Place a steel nut on the plate.
2. Hold the paper plate in one hand and a magnet under the paper plate with the other hand. Slowly move the magnet to drag the steel nut around through the paint. Use your imagination to create your own colourful designs.

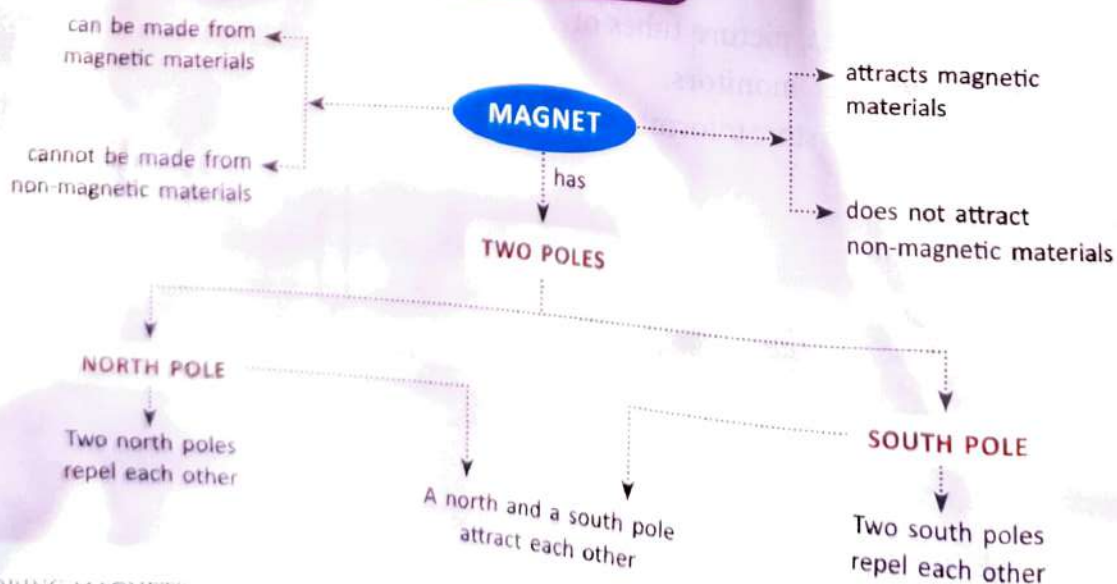
## Oral Questions for Class Discussion

1. Can a magnet be used to magnetize a piece of iron? How?
2. Why is it recommended that magnets be stored with keepers?
3. Is the influence of a magnet felt only near the poles or all around the magnet?
4. Name two gadgets used at home that have magnets.

## NOW YOU KNOW

- ❖ A magnet can attract iron and other magnetic materials.
- ❖ In a magnet, the regions of strongest magnetism are called poles.
- ❖ A magnet has two poles—north pole and south pole.
- ❖ The poles of a magnet always exist in pairs.
- ❖ In a freely suspended magnet, the north pole points towards the geographical north and the south pole towards the geographical south.
- ❖ The needle of a compass indicates the north–south direction.
- ❖ In magnets, unlike poles attract and like poles repel each other.
- ❖ The earth behaves like a huge magnet with its north pole near the geographical North Pole and its south pole near the geographical South Pole.

## CONCEPT MAP



## NEW WORDS

**MAGNETIC MATERIALS**—materials that are attracted by a magnet

**NORTH POLE**—the end of a magnet that points north when the magnet is freely suspended

**SOUTH POLE**—the end of a magnet that points south when the magnet is freely suspended



## EXERCISES

### A. Multiple-choice Questions: Choose the most appropriate option.

- Which of the following is not a magnetic material?
  - gold
  - nickel
  - iron
  - cobalt
- The magnetic strength of a magnet
  - is uniform throughout the body of the magnet.
  - is concentrated at one point called the pole of the magnet.
  - is concentrated in two regions called the poles of the magnet.
  - is concentrated in the centre of the magnet.
- A bar magnet is rolled in a heap of 18 steel pins. The number of pins sticking to the north pole (N), the south pole (S) and the centre (C) are counted. Which of the following is the most likely outcome?
  - N-2, S-8, C-8
  - N-8, S-2, C-8
  - N-8, S-8, C-2
  - N-6, S-6, C-6
- A sheet of thick paper was placed between a magnet and iron nails. It was found that the magnet could still attract the iron nails. This showed that:
  - Paper is a magnetic material.
  - Paper allows magnetic force to pass through it.
  - The paper was magnetized.
  - Paper does not allow magnetic force to pass through it.
- If the red end of each bar magnet is the north pole and the blue end is the south pole, which of the following arrangements of bar magnets is possible to make?
 

a.

b.

c.

d.
- Which of the following cannot happen, irrespective of whether the given iron bar B is a bar magnet or not?
  - Both ends of B are attracted to the north pole of a magnet.
  - Both ends of B are attracted to the south pole of a magnet.
  - One end of B is attracted and the other end is repelled by the north pole of a magnet.
  - Both ends of B are repelled by the south pole of a magnet.

7. Which of these materials can be used to make the needle of a compass?
  - a. copper
  - b. aluminium
  - c. steel
  - d. gold
8. A compass needle is kept in the centre of a bar magnet. Which of the following is true?
  - a. It will point in the north-south direction.
  - b. It will align itself along the length of the magnet with its N pole towards the N pole of the magnet.
  - c. It will not point in any specific direction.
  - d. It will align itself along the length of the magnet with its S pole towards the N pole of the magnet.

**B. Very Short-answer Questions: Fill in the blanks.**

1. Like poles repel and unlike poles attract each other.
2. Materials that are attracted by a magnet are called magnetic materials.
3. Compass is an instrument used to find directions, whose working is based on direction property of magnets.
4. Iron pieces used for storing magnets are called magnetic keepers.

**C. Short-answer Questions:**

**Answer in a sentence or two.**

1. Why does a freely suspended magnet point in the north-south direction?
2. What will happen to the poles of a magnet if it is broken into two?
3. Will a strong magnet attract a piece of copper? Why?
4. State the rule for attraction and repulsion between two magnets.
5. One precaution you should always take while handling a magnet is not to drop it. Why?
6. Why are magnetic keepers used to store magnets?

**Are these statements true or false? In each case give the reason for your opinion.**

7. A magnetic north pole cannot exist without a magnetic south pole.
8. The single touch method of making a magnet is widely used nowadays to make magnets.
9. If a magnet is heated, it can lose its magnetism.
10. Magnets can be made from magnetic or non-magnetic materials.

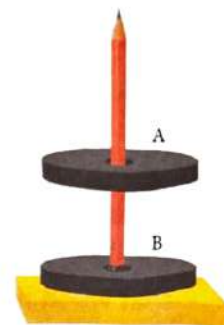
**D. Long-answer Questions: Answer in 30-40 words.**

1. How will you experimentally show that there are two regions in a magnet where its magnetic strength is concentrated?
2. Define: a. Magnetic force b. Magnetic poles
3. Give a step-by-step method to test if a piece of magnetic material is a magnet.
4. Name and describe the instrument that uses a magnet to find direction.
5. List any three uses of magnets.



### HOTS Questions: Think and answer.

1. Before invention of the compass, sailors used stars to find directions. In what way is using a compass better?
2. While screwing iron screws using a screwdriver, the screws often fall off, making it inconvenient. Suggest a method by which the screws can be prevented from falling off.
3. A and B are ring magnets. Explain why A remains suspended in air. Mark the N and S poles in both magnets. Can you suggest a method to make magnet A come in contact with magnet B without pushing it down?
4. Maglev train is a type of train that moves while floating in air just above its track. This way friction is minimized and the train moves very fast—at speeds exceeding 500 km per hour. How do you think it is made to float above the rails?

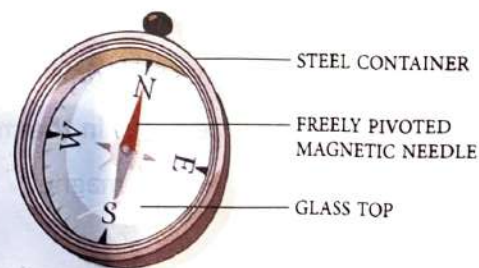


### BE A SCIENTIST

#### Problem-solving

A company that made scientific instruments designed a magnetic compass (shown here), and sold it in the market. However, customers complained that the compass did not show the direction correctly.

The company tested the compasses and found that the needle did not point in any one direction. On further testing, it was confirmed that the needle was properly magnetized and freely pivoted. They could not understand what was wrong. Can you?



## ENRICHMENT ACTIVITIES AND PROJECTS



### In the Library—Research Project

#### Inquiry-based learning; Collaboration

- ☐ Magnetic inks are made by mixing very small particles of a magnetic material with a liquid. The mixture finds several uses. Banks use magnetic inks on cheques. Machines can then be used to sort out the cheques automatically. The tapes that were earlier used in cassette recorders were made by coating a thin layer of magnetic ink on a plastic tape. Find out how the tapes recorded sound. Find out about other uses of magnetic inks.
- ☐ **Group project:** Magnets are used in many different fields. Find out about the various uses of magnets in the field of medicine.





## My Virtual Library—Research/Activities on Internet

- Visit [rsgr.in/nlsc-6](http://rsgr.in/nlsc-6) and click on LINK 9 for an interactive activity on magnets.
- Visit [rsgr.in/nlsc-6](http://rsgr.in/nlsc-6) and click on LINK 10 to learn more about magnets and do experiments on magnetism.
- Visit [rsgr.in/nlsc-6](http://rsgr.in/nlsc-6) and click on LINK 11 to make your own compass.



## To Meet—Research on Scientist

Inquiry-based

**Hans Christian Oersted** (1777–1851) In the year 1820, Hans Christian Oersted, a Danish professor was doing experiments with electricity. He noticed that when he passed an electric current through a wire, a compass needle lying near the wire showed a deflection and no longer pointed in the north–south direction. This made Oersted realise that an electric current can cause a magnetic effect. He had discovered what is now known as **electromagnetism**, that is, producing magnetism with an electric current.

Find out more about Oersted. Find out about the uses of electromagnets.



## In the Laboratory—Experimental Project

Experimental

### To study the differences in the magnetic properties of iron and steel

You will need ♦ a bar magnet ♦ an iron bar (soft iron) ♦ a steel bar of the same size ♦ iron filings

Procedure:

- ♦ Keep the iron filings in a pile on a piece of paper. Keep the soft iron bar over the filings. Touch the upper end of the iron bar with the bar magnet (Fig. 1). You will find that the iron filings now stick to the iron bar. This shows that the iron bar has become magnetized by the magnet. Lift up the magnet and the iron bar (Fig. 2). Remove the iron filings stuck to the iron bar and keep them separately. Make a similar pile of iron filings again, and repeat the experiment with the steel bar.

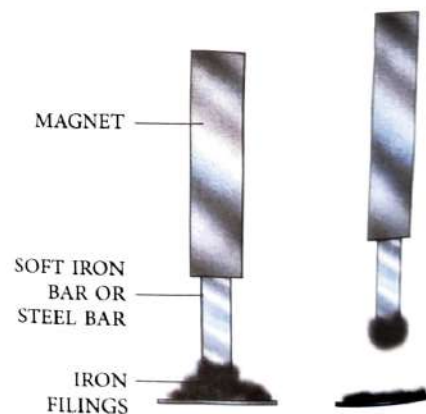


Fig. 1

Fig. 2

Compare the amount of iron filings picked up by the iron and steel bars. In which case do the iron filings more? Which gets magnetized more by touching with a magnet—iron or steel?

- ♦ If a magnetic material is magnetized by touching it with a magnet, it retains a part of its magnetism even after the magnet is removed. To test which retains more magnetism—iron or steel, perform the following experiment. Repeat the experiment described above at the point of lifting up the magnet and the bar. Now remove the magnet (Fig. 3). Most of the iron filings will fall off, but some will remain stuck to the bar. Compare the amount of iron filings that remain stuck to the iron bar and the steel bar. In which case are the iron filings more? Which retains more magnetism—iron or steel?

## NEW ASSESSMENT PATTERN

### ASSERTION-REASON TYPE QUESTIONS

The following questions are Assertion-Reason type questions. These consist of two statements—Assertion (A) and Reason (R). Answer these questions selecting the most appropriate option given below.

1. **Assertion (A):** Repulsion and not attraction is the sure test of magnetism.

**Reason (R):** Like poles of magnets attract and unlike poles repel.

- a. Both A and R are true, and R is the correct explanation of A.
- b. Both A and R are true, but R is not the correct explanation of A.
- c. A is true, but R is false.
- d. A is false, but R is true.



2. **Assertion (A):** The effect of a magnet is felt up to a certain distance only.

**Reason (R):** The influence of a magnet extends to a limited region around the magnet.

- a. Both A and R are true, and R is the correct explanation of A.
- b. Both A and R are true, but R is not the correct explanation of A.
- c. A is true, but R is false.
- d. A is false, but R is true.

### CASE-BASED QUESTIONS

Read the passage/data/case and answer the questions given below.

**Maglev trains:** What if you could travel from Srinagar to Kanniyakumari in just under six hours without boarding a plane? It could be possible on a Maglev train.

Maglev is short for magnetic levitation. Levitation means floating in air. Maglev trains use the repulsion property of magnets to actually float in air! Because of this, there is almost no friction, and the trains can run very fast—at several hundred kilometres an hour. Also, they run very smoothly, as there is far less noise and vibration than in typical trains.

The idea came to a scientist called James Powell as he sat in a traffic jam, thinking that there must be a better way to travel on land than cars or traditional trains. He dreamed up the idea of using very powerful magnets to levitate a train. James Powell and Gordon Danby of Brookhaven received the first patent for a magnetically levitated train design in the late 1960s. It took quite some time for the idea to be converted into reality. The first commercially operated high-speed Maglev train started in Shanghai in 2004. Other such trains are in operation in Japan and South Korea.

Choose the most appropriate option.

- 1. Which of these statements is true?
  - a. Maglev trains run very fast and smoothly.
  - b. Maglev trains are not in contact with the ground.
  - c. Both a and b
  - d. Neither a nor b
- 2. Which of these statements is false?
  - a. Maglev trains float in air.
  - b. Maglev trains are lighter than air.
  - c. Both a and b
  - d. Neither a nor b
- 3. Once James Powell made the design of the first magnetic levitation, it was quickly developed and brought into commercial use. Do you agree with this statement?
  - a. Yes
  - b. No
  - c. Cannot say